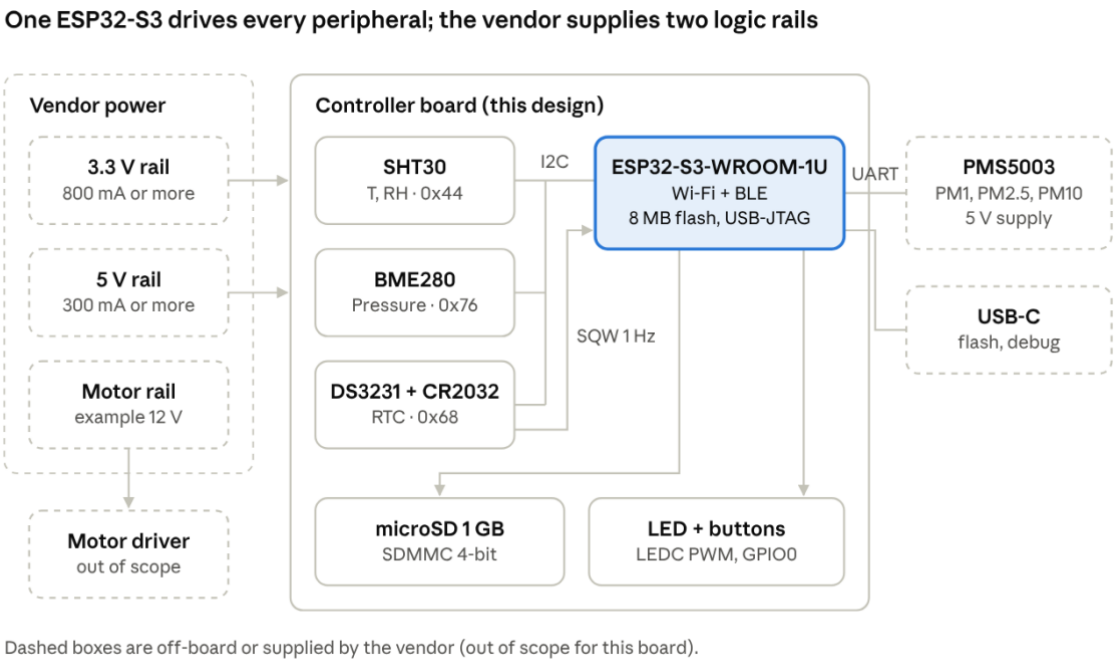
Hardware Requirement Sheet: Praan AQ Node

Controller board (MCU + peripherals). Power distribution, connectors and motor driver are vendor-supplied.

Block diagram



Dashed boxes are off-board or supplied by the vendor (out of scope for this board).

Component selection

Part	Function	Why chosen	Alternative
ESP32-S3-WROOM-1U-N8	MCU, Wi-Fi + BLE	Dual core, SDMMC host, 3 UARTs, native USB-JTAG, Secure Boot, pre-certified, U.FL antenna	ESP32-C6
SHT30-DIS-B	Temperature, humidity	±0.2 °C, low drift (< 0.25 %RH/yr), primary T/RH source	SHT31 (drop-in, ±2 %RH)
BME280	Pressure	±1 hPa; its T/RH used as fallback and drift check	BMP280
DS3231SN + CR2032	RTC, 1 Hz tick	±2 ppm TCXO, valid time offline, SQW paces sampling; no charge path	RV-3028-C7
microSD 1 GB (SDMMC)	Data storage	Store-first logging, ~1 year of backlog at 32 B/s	SPI NAND (smaller)
RGB LED + 2 buttons	Status, reset, provisioning	Field diagnostics; BOOT pin doubles as user button	WS2812
PMS5003 (off-board)	PM1 / PM2.5 / PM10	Specified by task; UART, 3.3 V logic	Sensirion SPS30

33 parts on the board. Full values and wiring: schematic.pdf.

Power requirements for the vendor

Rail	Loads	Required rating	Regulation
3.3 V	MCU, SD, sensors, LED	≥ 800 mA (468 mA peak × 1.5)	±3 %, < 50 mVpp, > 3.0 V on TX bursts
5 V	PMS5003	≥ 300 mA (150 mA peak × 1.5)	±5 % (PMS range 4.5–5.5 V)
Motor	2× drive motors	From motor datasheet (example: 12 V, ≥ 7.5 A peak)	Separate rail, star ground with logic

- **Power-good signal** (open-drain) to MCU IO38 with ≥ 50 ms hold-up on 3.3 V, so firmware can flush SD on power loss.
- **Motor noise** must not reach the 3.3 V rail; logic and motor returns meet at a single star point.
- Load breakdown and energy usage: power_budget.pdf